Teacher Wu Kuan | Digital SAT Math — Key English Phrases Companion

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1. The trap question

Many students can do the math but get stuck on the English wording, e.g. “The system has exactly one real solution.”

2. The method

• “exactly one real solution” (quadratic) → discriminant $b^2 - 4ac = 0$.
• “no solution” (linear system) → same slope, different intercept (parallel lines).
• “infinitely many solutions” → the two equations are the same line (all coefficients proportional).
• “y-intercept” → the value of y when $x = 0$.
• Translate words to symbols first, then solve.

3. Parent check method

Have your child underline the key phrase and say in their own words what it means before calculating.

Practice

1. $x^2 + 6x + c = 0$ has exactly one real solution. Find c.
2. The system $2x + 3y = 5$ and $4x + 6y = k$ has no solution. What values can k take?
3. The system $3x - y = 4$ and $6x - 2y = c$ has infinitely many solutions. Find c.
4. What is the y-intercept of $y = 4x - 7$?
5. “Twice a number decreased by 5 is 11.” Write an equation and solve.

Answers

1. $c = 9$ ($36 - 4c = 0$)
2. Any $k \neq 10$
3. $c = 8$
4. $(0, -7)$
5. $2n - 5 = 11$, $n = 8$

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